SEMM Primer - Ordinary Differential Equations (ODE)

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August 20, 2026

Goals of the session

By the end of this session you'll be able to:

- Identify and distinguish the different types of differential equations.
- Connect them to the physical phenomena they model (1 example).
- Understand solution methods for ODEs commonly encountered in SEMM (3 methods).

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What is a differential equation?

An equation that involves a function and its derivatives.

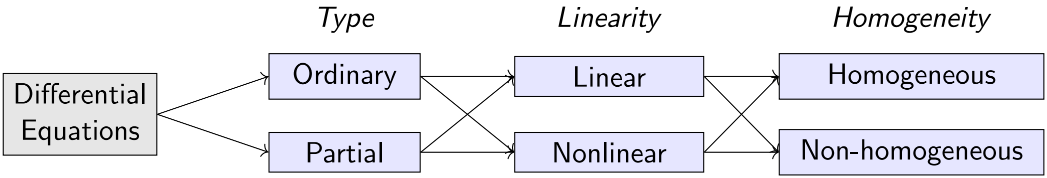
Some examples:

$$\frac{dy}{dx} - 2y = x,$$

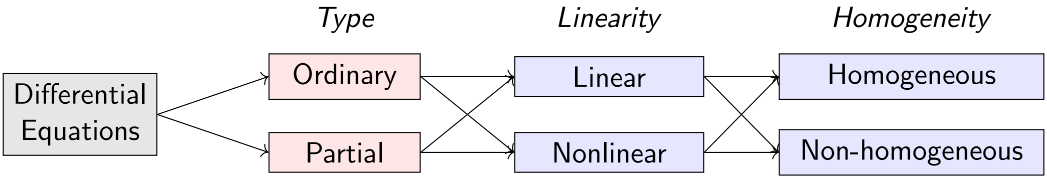
$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = \sin x,$$

$$\frac{\partial^2 u}{\partial t^2} + 2\frac{\partial u}{\partial t} = \frac{\partial^2 u}{dx^2}$$

Categories of differential equations



Categories of differential equations



Ordinary

$$\frac{dx}{dt} = kx, \quad x(t)$$

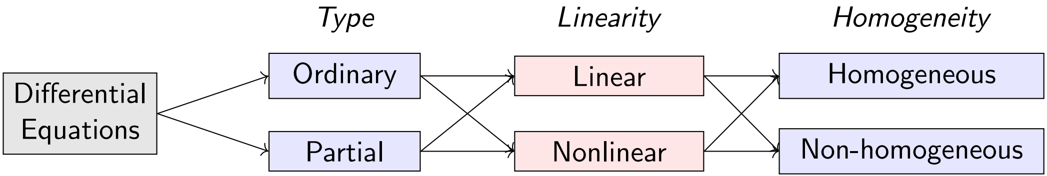
$$e^2 \frac{d^2y}{dx^2} + \sin x \frac{dy}{dx} + x^2 y = \frac{1}{x}, \quad y(x)$$

Partial

$$\frac{\partial y}{\partial t} + c\frac{\partial y}{\partial x} = 0, \quad y(t,x)$$

$$\frac{\partial y}{\partial t} + y\frac{\partial y}{\partial x} = \nu \frac{\partial^2 y}{\partial x^2}, \quad y(t,x)$$

Classes of Differential Equations



Linear

$$\frac{dx}{dt} = t^2, \quad x(t)$$

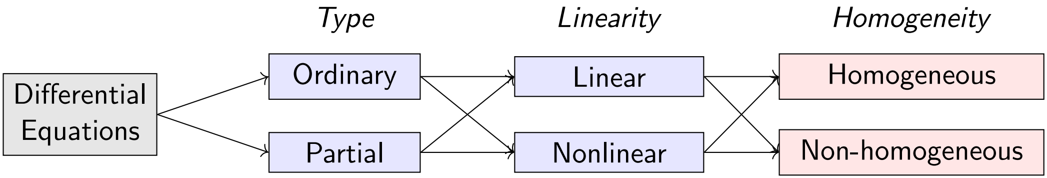
$$e^x \frac{d^2y}{dx^2} + \sin x \frac{dy}{dx} + x^2 y = \frac{1}{x}, \quad y(x)$$

Nonlinear

$$\frac{dx}{dt} = x^2, \quad x(t)$$

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\sin\theta = 0, \quad \theta(t)$$

Classes of Differential Equations



Homogeneous

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = 0, \quad x(t)$$

$$\frac{dx}{dt} = x^2, \quad x(t)$$

Non-homogeneous

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = \sin(2\pi t), \quad x(t)$$

$$\frac{dx}{dt} = t^2, \quad x(t)$$

Contents

Some general concepts that you must remember

Let's consider a general ODE to be:

$$F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, ..., \frac{d^ny}{dx^n}\right) = 0$$

- The term *ordinary* refers to the fact that it deals with functions of a single variable ($y(x)$ as opposed to $y(x_1, x_2, ...)$.
- The *order* of the ODE is the highest derivative appearing.
- A *solution* of an ODE is a function $\hat{y}(x)$ that satisfies the ODE.
- In finding a solution, we generally find a family of solutions, with a set of undetermined coefficients.
- These coefficients are found after we are given extra information in the form of *boundary conditions* (BVP) or *initial conditions* (IVP).
- Let's start with a full worked example, from first principles to the particular solution.

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Worked example: suspension bridge

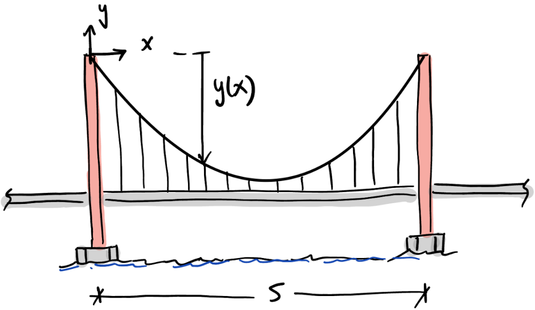
Let's motivate our study of ODEs with a basic example: the **shape of the cable** in a suspension bridge.

Figure: The Golden Gate Bridge

Let's quickly go over this example, from top to bottom.

Worked example: suspension bridge

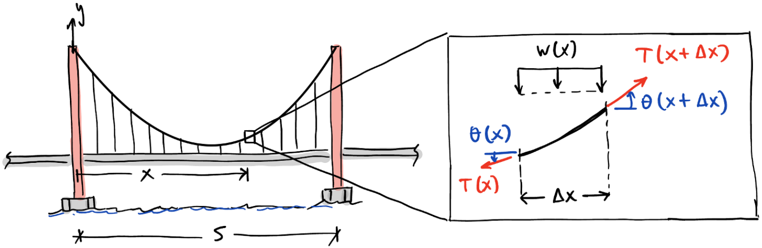
This is a little drawing of the bridge, with some parameters we'll use for the solution.



To find the governing equation, we can use a fundamental law: equilibrium!

Example: the suspension bridge

A free body diagram of a portion of the cable:



Now, let's use our fundamental law. From horizontal equilibrium:

$$-T(x)\cos(\theta(x)) + T(x+\Delta x)\cos(\theta(x+\Delta x)) = 0$$

$$\frac{T(x+\Delta x)\cos(\theta(x+\Delta x)) - T(x)\cos(\theta(x))}{\Delta x} = 0$$

Example: the suspension bridge

$$\frac{T(x+\Delta x)\cos\left(\theta(x+\Delta x)\right) - T(x)\cos\left(\theta(x)\right)}{\Delta x} = 0$$

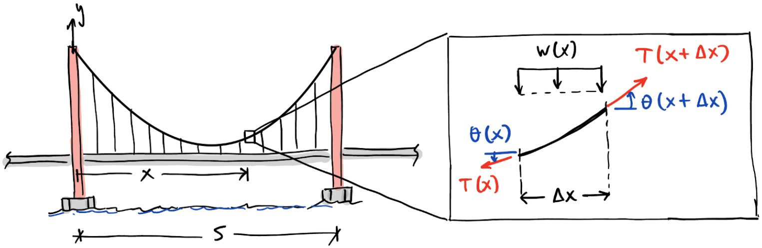
From horizontal equilibrium:

$$\lim \Delta x \to 0 \Rightarrow (T(x)\cos\theta(x))' = 0$$

$$\Rightarrow T(x)\cos\theta(x) = H$$

So, the horizontal component of the tension in the cable is constant.

Example: suspension bridge



From the vertical equilibrium equation we find:

$$\frac{T(x+\Delta x)\sin(\theta(x+\Delta x)) - T(x)\sin(\theta(x))}{\Delta x} = w(x)$$

Example: suspension bridge

$$\frac{T(x+\Delta x)\sin(\theta(x+\Delta x)) - T(x)\sin(\theta(x))}{\Delta x} = w(x)$$

$$\lim \Delta x \to 0 \Rightarrow \frac{d}{dx}(T(x)\sin(\theta(x))) = w(x)$$

But $T(x)\cos(\theta(x)) = H$, so:

$$\frac{d}{dx}\left(H\frac{\sin\theta(x)}{\cos\theta(x)}\right) = w(x)$$

$$\frac{d}{dx}(\tan\theta(x)) = \frac{w(x)}{H}$$

If θ is small, $\tan\theta \approx \theta \approx dy/dx$:

$$\boxed{\frac{d^2y}{dx^2} = \frac{w(x)}{H}}$$

Example: suspension bridge

Our governing equation is:

$$\boxed{\frac{d^2y}{dx^2} = \frac{w(x)}{H}}$$

This is what we call a "separable ODE":

$$d^2y = \frac{w(x)}{H}dx^2$$

To solve, integrate twice on each side, and add integration constants. Let's assume $w(x) = w$ (constant).

$$dy = \frac{w}{H}xdx + C_1$$

$$y(x) = \frac{w}{2H}x^2 + C_1x + C_2$$

Example: suspension bridge

How to find C_1 and C_2?

Summary

- We start with a sensible **model** of reality.
- We may want to make some **assumptions** to simplify the math.
- We establish **relations** between the variables at play using a fundamental law (e.g., equilibrium).
- We do some **algebra** to find a differential equation that describes the behavior of the system.
- We **solve** the differential equation to find a general solution.
- We enforce the boundary/initial conditions to find a unique solution to our specific problem (if it exists!)

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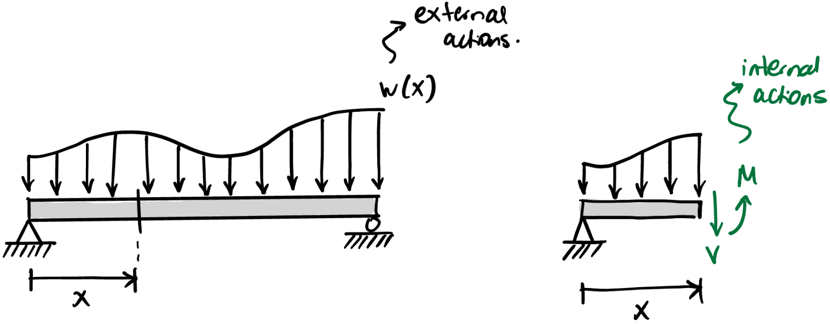
Relevant ODEs in SEMM

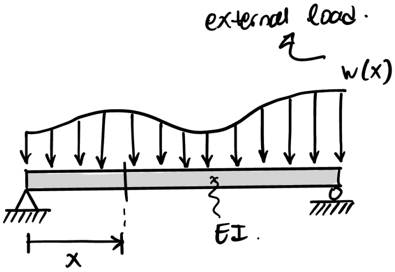
The internal moment $M(x)$ of a horizontal beam subjected to a distributed load $w(x)$:

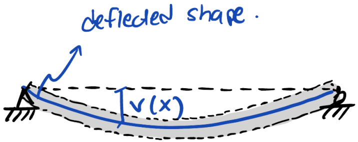
$$\frac{d^2M(x)}{dx^2} = w(x)$$



Relevant ODEs in SEMM

The Euler-Bernoulli beam equation for the deflections $v(x)$:

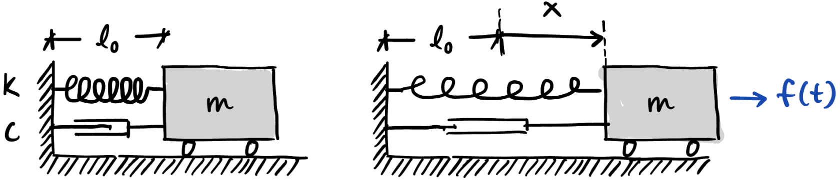




$$\frac{d^2}{dx^2}\left(EI\frac{d^2v}{dx^2}\right) = w(x)$$

Relevant ODEs in SEMM

The position $x(t)$ of a mass-spring-damper system subjected to an external force $f(t)$:



$$m\frac{d^2x(t)}{dt^2} + c\frac{dx(t)}{dt} + kx(t) = f(t)$$

Relevant ODEs in SEMM

Summary:

Physical phenomenon	Equation	Order
Suspension bridge cable	$\frac{dy^2}{dx^2} = \frac{w(x)}{H}$	Second order
Bending moment in a beam	$\frac{d^2M(x)}{dx^2} = w(x)$	Second order
Deflections in a beam	$\frac{d^2}{dx^2}\left(EI\frac{d^2v}{dx^2}\right) = w(x)$	Fourth order
Dynamics of mass-spring-dashpot system	$m\ddot{x} + c\dot{x} + kx = f(t)$	Second order

Table: Common ODEs in SEMM

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Solving linear ODEs

A **linear** ODE:

$$\boxed{a_n(x)\frac{d^n y}{dx^n} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)}$$

If the coefficients are independent from x, then we call it a linear ODE with constant coefficients:

$$\boxed{c_n\frac{d^n y}{dx^n} + \cdots + c_1\frac{dy}{dx} + c_0 y = g(x)}$$

A counter-example... For instance:

$$\ddot{\theta} + \frac{g}{L}\sin\theta = 0$$

is a nonlinear ODE, and models a simple pendulum of length L.

Solving linear ODEs

For linear systems, the principle of superposition applies.

Solving linear ODEs

If we want to solve:

$$\boxed{a_n(x)\frac{d^n y}{dx^n} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)}$$

we can:

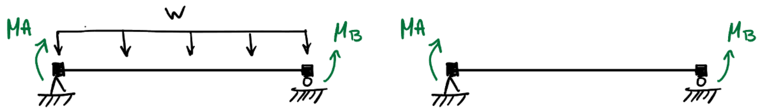
- Solve the homogeneous version of the equation.

  $$a_n(x)\frac{d^n y}{dx^n} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = 0$$

  We'll call this the complementary solution $y_h(x)$.
- Then, we find a "particular solution" $y_p(x)$ to the non-homogeneous version of the equation.
- We find the general solution as the superposition of the complementary and the particular solutions:

$$\boxed{y(x) = y_h(x) + y_p(x)}$$

Example: internal moment in a beam



The governing equation (trust me for now):

$$\frac{d^2M}{dx^2} = w$$

And the homogeneous version:

$$\frac{d^2M}{dx^2} = 0$$

We can integrate twice to find a solution:

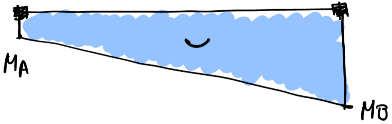
$$\frac{d^2M}{dx^2} = 0 \Rightarrow \frac{dM}{dx} = c_1$$
$$\Rightarrow M(x) = c_1x + c_2$$

The constants c_1 and c_2 are computed from the "boundary conditions" of the problem. Let's add the subscript:

$$M_h(x) = c_1x + c_2$$

If we apply the boundary conditions here:

$$M_h(x) = \left(\frac{M_B - M_A}{L}\right)x + M_A$$



Example: internal moment in a beam

Then, we find a particular solution to our equation. Let's consider the case where $w(x)$ is constant (w). We can easily see that:

$$M_p(x) = d_1 x + d_2 + \frac{1}{2} w x^2$$

represents a set of possible solutions to our ODE, with d_1 and d_2 constants.
The general solution will be:

$$M(x) = M_h(x) + M_p(x) = (c_1 + d_1)x + (c_2 + d_2) + \frac{1}{2} w x^2$$

You can check that this one is a particular solution if you plug it back into the original ODE we are trying to solve.

Example: internal moment in a beam

Our particular solution also has some constants, which we will compute using the zero boundary conditions (we've already applied the end moments in the homogeneous solution, so we can't add them twice!).

$$M_p(x) = d_1 x + d_2 + \frac{1}{2} w x^2$$

If we apply $M_p(0) = 0$ and $M_p(L) = 0$ we find:

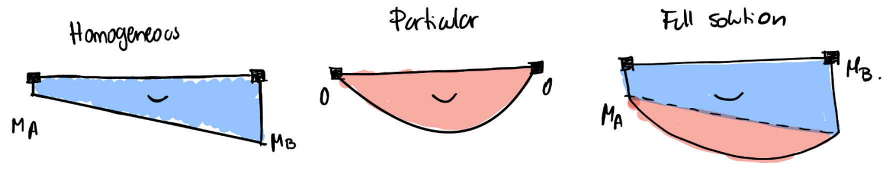
$$M_p(x) = -\frac{wLx}{2} + \frac{wx^2}{2}$$
$$M_p(x) = \frac{w}{2}\left(x^2 - xL\right)$$

Example: internal moment in a beam

Thus, the general solution is:

$$\boxed{M(x) = \left(\frac{M_B - M_A}{L}\right)x + M_A + \frac{w}{2}\left(x^2 - xL\right)}$$

And this looks like:



Example: mechanical vibrations

Let's solve the linear ODE with constant coefficients:

$$A\ddot{x}(t) + B\dot{x}(t) + Cx(t) = f(t)$$

Following the same strategy, we begin by solving the homogeneous version, and then add a particular solution.

Example: mechanical vibrations

The homogeneous version looks like:

$$A\ddot{x} + B\dot{x} + Cx = 0$$

An good guess for the solution is:

$$x(t) = e^{rt}$$

If we plug it in:

$$Ar^2e^{rt} + Bre^{rt} + ke^{rt} = 0$$

We want a non-trivial solution, so $e^{rt} \neq 0$, and:

$$Ar^2 + Br + C = 0$$

Example: mechanical vibrations

The polynomial:

$$Ar^2 + Br + C = 0$$

is called the “characteristic equation” of our LODEwCC.
Solving the roots of this polynomial yields:

$$r_1, r_2 = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

Example: mechanical vibrations

In general, we get three possible scenarios out of this:

$$r_1, r_2 = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

- Scenario 1: two distinct and real roots (when $B^2 - 4AC > 0$).
- Scenario 2: two distinct complex roots (when $B^2 - 4AC < 0$).
- Scenario 3: a single real root (when $B^2 - 4AC = 0$).

Example: mechanical vibrations

Scenario 1: r_1 and r_2 are two distinct real roots.

$$x_c(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

And c_1 and c_2 can be found from initial conditions.

Example: mechanical vibrations

Scenario 2: r_1 and r_2 are two distinct imaginary roots, on the form:

$$r_1, r_2 = \lambda \pm i\mu$$

Thus:

$$x_c(t) = c_1 e^{(\lambda + i\mu)t} + c_2 e^{(\lambda - i\mu)t}$$

Which can be rewritten (using Euler's formula[1]) as:

$$x_c(t) = e^{\lambda t}(c_3 \cos \mu t + c_4 \sin \mu t)$$

[1] $e^{i\mu t} = \cos \mu t + i \sin \mu t$

Example: mechanical vibrations

Scenario 3: $r_1 = r_2 = -\dfrac{B}{2A}$

We could say:

$$x_c(t) = c_1 e^{rt} + c_2 e^{rt} = c_3 e^{rt}$$

However, this is not general enough to solve for two initial conditions (for $x(0)$ and $\dot{x}(0)$.
A more general guess would be:

$$d(t)e^{rt}$$

Plugging this one into our ODE yields:

$$e^{rt}(\ddot{d}(t)) = 0$$

Thus,

$$x_c(t) = (c_4 + c_5 t)e^{rt}$$

Example: mechanical vibrations

Let's now go back to our full ODE:

$$A\ddot{x} + B\dot{x} + Cx = f(t)$$

The solution to this inhomogeneous ODE will be:

$$x(t) = x_c(t) + x_p(t)$$

Where $x_p(t)$ is called a "particular solution" to our inhomogeneous ODE.
The particular solution depends on the right-hand-side. Let's see what happens if the RHS is a sinusoidal function.

Example: mechanical vibrations

$$A\ddot{x} + B\dot{x} + Cx = \sin \alpha t$$

An educated guess for a particular solution is:

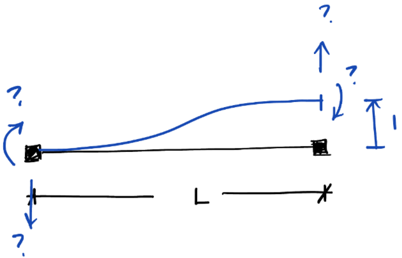
$$x_p(t) = C_1 \cos \alpha t + C_2 \sin \alpha t$$

We can solve the coefficients C_1 and C_2 plugging this guess (and its derivatives) into our inhomogeneous ODE.

Note

Comment

Now you can derive the stiffness coefficients for a frame element! (Check that out in the SEMM Primer).



$$\boxed{EI \cdot \frac{d^4 v}{dx^4} = 0}$$

Contents

Some useful books

- Krantz, Steven. *"Differential Equations"*. CRC Press. Available at UC Library.
- Duffy, Dean G. *"Advanced Engineering Mathematics with MATLAB". * CRC Press. Available online.
- Kreyszig, Erwin. *"Advanced Engineering Mathematics*. Wiley. Available at UC Library.

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